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Disease Trajectories from Healthcare Data: Methodologies, Key Results, and Future Perspectives

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Keywords

disease trajectories, healthcare data, multimorbidity, disease co-occurrence

Abstract

Disease trajectories, defined as sequential, directional disease associations, have become an intense research field driven by the availability of electronic population-wide healthcare data and sufficient computational power. Here, we provide an overview of disease trajectory studies with a focus on European work, including ontologies used as well as computational methodologies for the construction of disease trajectories. We also discuss different applications of disease trajectories from descriptive risk identification to disease progression, patient stratification, and personalized predictions using machine learning. We describe challenges and opportunities in the area that eventually will benefit from initiatives such as the European Health Data Space, which, with time, will make it possible to analyze data from cohorts comprising hundreds of millions of patients.

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INTRODUCTION

Unveiling Disease Co-Occurrence and Trajectories

It is well known that many chronic diseases tend to co-occur more than expected by chance, often as a consequence of shared risk factors, pathogenicity, or their treatment (1–5). In the literature the concepts of comorbidity and multimorbidity are not used systematically, although some papers aim for more clear distinctions (6, 7). The co-occurrence of diseases has attracted substantial attention during the last decade, facilitated by the ease of extracting structured diagnostic information from the electronic health records (EHRs) of millions of patients (7, 8). Increasing computational power and access to population-wide healthcare data have jointly paved the way for conducting studies by means of disease trajectories. Pioneers discussing the concept of disease trajectories include Ohlsson et al. (9) as well as Hyman & Corbin (10), who described disease trajectories as the course or progression of chronic disease presenting over time.

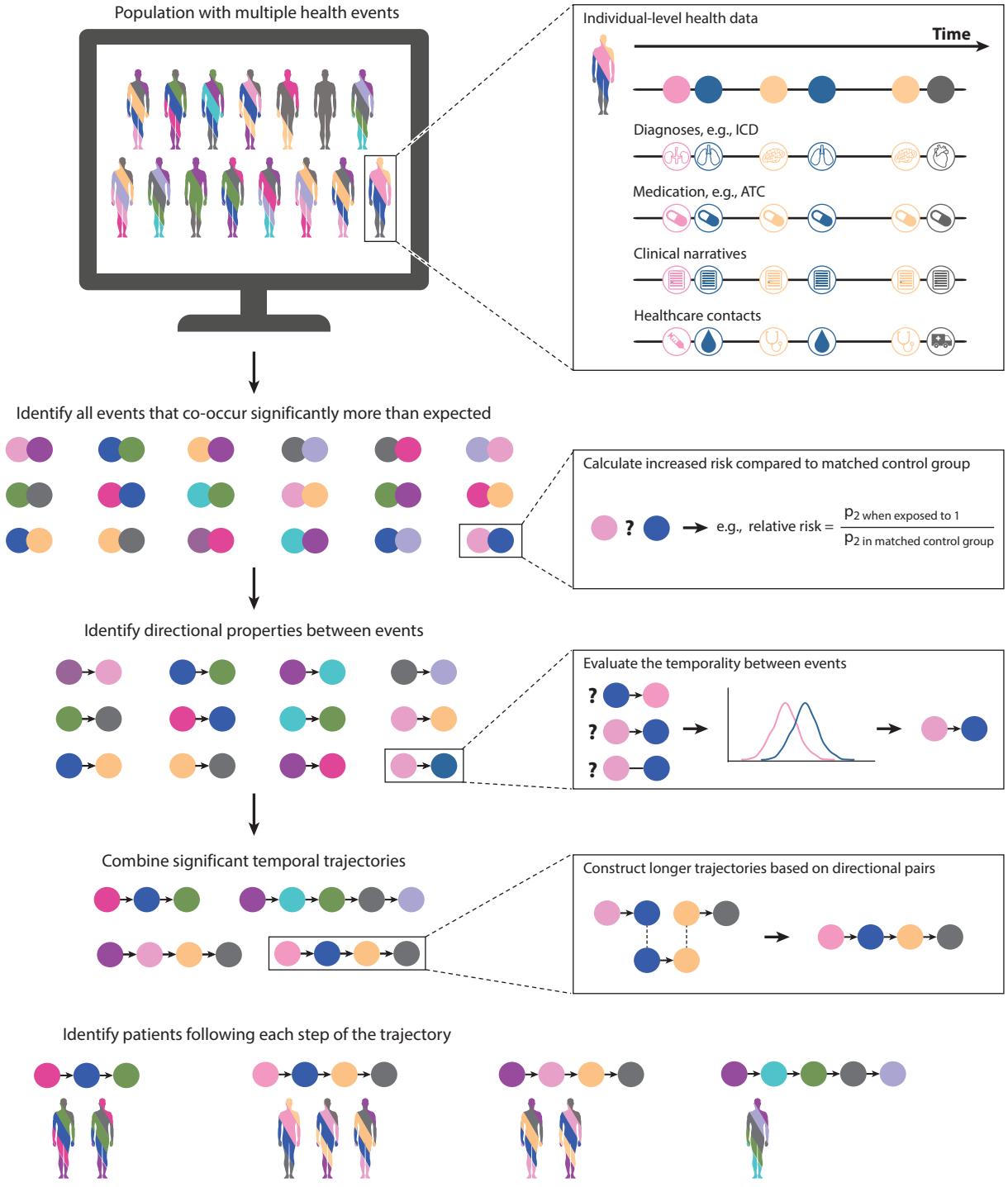
Temporal disease trajectories have the potential to contribute to the understanding of the causes and course of health events over time, which can lead to the identification of individuals at risk, discovery of factors impacting health changes, and evaluation of intervention effects (11). Disease trajectories have the potential to uncover pathophysiological similarities and differences that ultimately will facilitate personalized risk prediction and targeted treatment (12). For example, heart failure represents a symptom of a disease, a disease in itself, and a disease complication. Today, however, the treatment is often the same in all three cases, though it needs more systematic and contextual handling through precision medicine strategies (12).

Since the inception of disease trajectories, several studies have focused on describing diagnosis trajectories in a data-driven manner, mainly using repositories describing disease progression over time—typically national registries or EHR datasets of considerable size. **Figure 1** gives an overview of some common data types from EHRs that can be used for trajectory analyses. In principle, trajectory approaches can be used to investigate any kind of temporal health data, including categorical variables like encoded diagnoses, procedures, drugs, or drug shifts, as well as continuous features such as laboratory test or drug dosage data. More recent research, for example, has focused on prescription trajectories, using data on drug dosage changes as well as other events extracted from clinical notes (13–15).

Temporality

The main difference between diseases that co-occur and directional disease trajectories is the temporal aspect. Two diseases can co-occur without any specific direction, but for disease trajectories, one disease needs to occur significantly before the other (16). Hence, a summary-level disease trajectory highlights the sequence of diseases over the life course followed by numerous individual patients that share a similar temporal pattern, as visualized in **Figure 1**.

When millions of patients are analyzed, specific longitudinal subgroups may be identified that differ in how patients reach a given diagnosis. Hypothesis-generating biological information can be extracted not only from the co-occurrence of diseases but also from the directional relationships between diseases. Empirical evidence has demonstrated that the sequence in which diseases manifest influences the overall burden of comorbidity, as exemplified by the fact that patients with diabetes prior to sleep apnea carry a greater disease burden than those diagnosed with sleep apnea before diabetes (17). Furthermore, the direction between diseases can change depending on, for example, sex, where opposite temporal associations can be found in men and women for certain pairs of diseases (18). Additionally, prediction studies have robustly shown that using the temporal order of diseases outperforms using only the diagnoses themselves as a bag of words, for example, when predicting mortality of patients admitted to intensive care units



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Figure 1 (*Figure appears on preceding page*)

Overview of the creation of trajectories. A population with multiple health events is defined. Here each color represents a different event. Each individual should have one or more types of sequential health data available. The health data can be diagnoses, medications, events from clinical narratives, healthcare contacts, other types of sequential information, or combinations thereof. To create trajectories, events that co-occur significantly more than expected are identified. This can be based on several different metrics, such as the relative risk included here. Additional metrics are included in **Table 1** and discussed in the main text. Subsequently, directional properties between events are identified to discover which event happens significantly earlier than the other event. Trajectories of varying lengths can be constructed based on the directional disease pairs. Additionally, patients following specific trajectories can be identified for subsequent analysis, or trajectories can be summarized in several ways, as visualized in **Figure 3**. Abbreviations: ATC, Anatomical Therapeutic Chemical Classification System; ICD, International Classification of Diseases. Elements in the figure were adapted from Siggaard et al. 2020 (63) (CC BY 4.0)

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(19) or when predicting pancreatic cancer 3 to 60 months into the future (20). As the directional relationships between diseases add information, disease trajectories may have higher prediction power, because they not only reflect germline genetic risks but also incorporate the lifestyle effect of exposures such as drugs, nutrition, and pollution over time. Additional socioeconomic factors that affect disease outcomes may be implicitly included, for example, educational ascertainment, patient adherence, or other factors. All together, these factors help increase our understanding of disease progression, allowing for longitudinal patient stratification and prediction of future outcomes like comorbidity burden and survival (8).

The Increasing Burden of Comorbidities

The interest in disease co-occurrences and trajectories is also justified by the increasing burden of co- and multimorbidity in aging populations (21). Over the last decades, more people are living with chronic diseases, and many patients end up dealing with not just one, but multiple health conditions at the same time (22–25). In some countries almost 50% of adults are diagnosed with more than one chronic disease, and there is no indication that number is decreasing (2, 24). The increase in the number of individuals with multiple chronic conditions is partly due to advances in the medical sciences and more health-promoting environments that increase the life span, leading to increased aging across populations (22, 23, 26). Furthermore, being diagnosed with one disease increases the likelihood of being diagnosed with additional diseases (27). Multimorbidity is associated with poorer quality of life (28), higher health service costs (29), and increased mortality (30); thus, studies that focus on multimorbidity development over time are essential (31).

In this review, we present and discuss studies of disease trajectories, highlighting differences in approaches and main results, as well as new opportunities and challenges. Our focus is on disease trajectories comprising numerous, different, categorical medical events, encompassing diagnoses, medications, procedures, healthcare contacts, etc. Hence, trajectories presenting changes in one quantitative measure (e.g., glucose levels) are not the main focus of this review. We focus on studies conducted on European populations or studies comparing trajectories of Europeans with other populations.

OVERVIEW OF DISEASE TRAJECTORY WORK BASED ON EUROPEAN DATA

In the following we give an overview of selected disease trajectory studies from Europe based on the definition provided above. A previous review on trajectories of multimorbidity discovered that most multimorbidity trajectory studies are based on data from European countries (31). This is not surprising given the one-payer organization of most European healthcare systems. **Table 1** and **Figure 2** provide an overview of selected published work within the field of disease trajectories annotated with the main focus area of each study and a few keywords on the method applied.

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Table 1 Overview of selected large-scale studies of disease trajectories

Authors	Year	Country	Sample size	Data source(s)	Follow-up (years)	Focus	Method(s)
Jensen et al. (16)	2014	Denmark	6,200,000	Diagnoses: secondary care	1996–2010 (14.9)	Clusters of disease trajectories	Relative risk, p-value, and binomial test
Beck et al. (59)	2016	Denmark	120,000	Diagnoses: secondary care	1996–2014 (18)	Disease trajectories leading to sepsis	Relative risk, p-value, and binomial test
Kannan et al. (27)	2016	Sweden	5,512,469	Diagnoses: secondary care	1997–2009 (13)	Clusters of diseases	Causal information fraction measure
Beck et al. (17)	2017	Denmark	90,157	Diagnoses: secondary care	1994–2015 (22)	Disease trajectories of sleep apnea	Relative risk, p-value, and binomial test
Jensen et al. (36)	2017	Norway	7,741	Clinical narratives	1999–2014 (16)	Cancer, disease, and event trajectories	Frequent item set mining
Giannoula et al. (53)	2018	Spain	643,358	Diagnoses: secondary care	2004–2011 (7)	Clusters of disease trajectories	Pairwise comparisons of temporal sequence of multimorbidity
Ashworth et al. (32)	2019	United Kingdom	5,597	Diagnoses: primary care	2004–2018 (15)	Disease trajectories for selected long-term conditions	Temporal sequence of multimorbidity patterns
Hassaine et al. (48)	2020	United Kingdom	2,204,178	Diagnoses: primary and secondary care	1985–2014 (29)	Clusters of disease trajectories	Nonnegative matrix factorization
Hu et al. (122)	2019	Denmark	700,000	Diagnoses: secondary care	1994–2015 (22)	Disease trajectories for 17 different cancer types	Relative risk, p-value, and binomial test
Jørgensen et al. (102)	2020	Denmark	73,213	Diagnoses: secondary care	1994–2016 (23)	Disease trajectories for vascular and Alzheimer's dementia	Relative risk, p-value, and binomial test
Lademann et al. (90)	2019	Denmark	6,600,000	Diagnoses: secondary care	1996–2014 (18)	Stratification of disease trajectories using symptoms	Relative risk, p-value, and binomial test
Nielsen et al. (19)	2019	Denmark	8,871	Diagnoses: secondary care	1994–2016 (23)	Disease trajectories to predict mortality in the intensive care unit	Temporal sequence of multimorbidity patterns
Westergaard et al. (18)	2019	Denmark	6,909,676	Diagnoses: secondary care	1994–2015 (22)	Sex-stratified, population-wide disease trajectories	Hierarchical Bayesian Poisson model and Mann-Whitney test
Aguado et al. (47)	2020	Spain	3,135,948	Diagnoses: primary care	2006–2017 (12)	Patients with two or more chronic conditions and subanalysis of type 2 diabetes mellitus	Logistic regression models with odd ratios

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Table 1 (*Continued*)

Authors	Year	Country	Sample size	Data source(s)	Follow-up (years)	Focus	Method(s)
Haug et al. (52)	2020	Austria	5,112,811	Diagnoses: secondary care	1997–2014 (18)	Clusters of disease trajectories	Temporal sequence of multimorbidity patterns
Kjærsgaard Andersen et al. (84)	2020	Denmark	14,488	Diagnoses: secondary care	1994–2018 (24)	Disease trajectories for hidradenitis suppurativa	Relative risk, p-value, and binomial test
Ploner et al. (39)	2020	Germany	7,466	Claims data: diagnoses, medication, and procedures	2011–2015 (5)	Health insurance claim trajectories for chronic obstructive pulmonary disease severity	Gradient boosting and logistic regression models
Siggaard et al. (63)	2020	Denmark	7,186,865	Diagnoses: secondary care	1994–2018 (24)	Population-wide disease trajectories	Relative risk, p-value, and binomial test
Aguayo-Orozco et al. (13)	2021	Denmark	7,255,919	Prescriptions	1995–2019 (24)	Population-wide prescription trajectories	Poisson regression model and binomial test
Han et al. (54)	2021	UK	144,496	Diagnoses: secondary care	1997–2019 (23)	Disease trajectories for depression	Conditional Cox regression, binomial test, and conditional logistic regression
Jørgensen & Brunak (88)	2021	Denmark	284,154	Diagnoses: secondary care	1994–2015 (23)	Disease trajectories to identify misdiagnosis	Relative risk, p-value, and binomial test
Shi et al. (58)	2021	Belgium	65,939	Diagnoses: primary care	1991–2015 (25)	Disease trajectories for adults over age 40	Markov chains and weighted association rule mining
Haue et al. (64)	2022	Denmark	570,157	Diagnoses: secondary care	1994–2018 (24)	Disease trajectories for ischemic heart disease	Relative risk, p-value, and multiple linear regression models
Henning et al. (83)	2022	Denmark	8,758	Diagnoses: secondary care	1994–2018 (24)	Disease trajectories for hyperhidrosis	Relative risk, p-value, and binomial test
Hou et al. (61)	2022	UK	395,539	Diagnoses: secondary care	1997–2020 (24)	Disease trajectories associated with coffee consumption	Relative risk, Pearson correlation, binomial test, and logistic regression
Künnapuu et al. (40)	2022	Estonia	147,000	Diagnoses: secondary care, prescriptions, and claims	2012–2019 (8)	General disease trajectories and type 2 diabetes mellitus trajectories	Relative risk, p-value, and binomial test

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Table 1 (*Continued*)

Authors	Year	Country	Sample size	Data source(s)	Follow-up (years)	Focus	Method(s)
Thygesen et al. (34)	2022	UK	56,945,027	Healthcare contacts	2020–2021 (2)	COVID-19 event trajectories	Temporal sequence of phenotypic COVID-19 events
Yang et al. (38)	2022	UK	502,650	Diagnoses: secondary care and biomarkers	1997–2019 (23)	Disease trajectory network and biomarker–disease trajectories with higher risk in women	Conditional logistic regression model using odds ratios, p-values, and binomial test
Dervić et al. (60)	2024	Austria	3,955,028	Diagnoses: secondary care	2003–2014 (11)	Sex- and age-stratified multilayered comorbidity networks	Relative risk, p-value, and binomial test
Hagmann et al. (56)	2023	Switzerland	257,164	Diagnoses: secondary care	2002–2018 (17)	Sex-stratified disease trajectories prior to chronic obstructive pulmonary disease	Conditional logistic regression and between-class principal component analysis
Hjaltelin et al. (37)	2023	Denmark	48,598	Diagnoses: secondary care and clinical narratives	2006–2016 (11)	Disease trajectories and symptom trajectories from clinical narratives for pancreas cancer patients	Relative risk, p-value, and binomial test
Jia et al. (62)	2023	UK	326,606	Diagnoses: secondary care	2006–2022 (17)	Metabolic dysfunction–associated fatty liver disease trajectories	Cox regression analysis, hazard ratio, p-value, and binomial test
Placido et al. (20)	2023	Denmark	6,200,000	Diagnoses: secondary care	1977–2018 (42)	Prediction of pancreas cancer using disease trajectories	Temporal sequence of multimorbidity patterns
Giannoula et al. (89)	2023	Spain	6,214	Healthcare contacts	2012–2016 (4)	Care trajectories of long-term breast cancer survivors	Dynamic time warping–based unsupervised clustering
Xia et al. (41)	2024	UK	344,609	Diagnoses: secondary care	1997–2019 (23)	Disease trajectories of individuals with high neuroticism	Phenome-wide association analysis with Cox regression and binomial test
Jørgensen et al. (33)	2024	Denmark	5,752	Diagnoses: secondary care	1977–2018 (42)	Clustering of disease trajectories for kidney transplant recipients	Relative risk, p-value, and binomial test
Monero et al. (57)	2024	Scotland	12,009	Diagnoses: primary care	2000–2006 (7)	Sex stratified multimorbidity patterns in the oldest (>90 years old)	Bayesian framework and association by chance

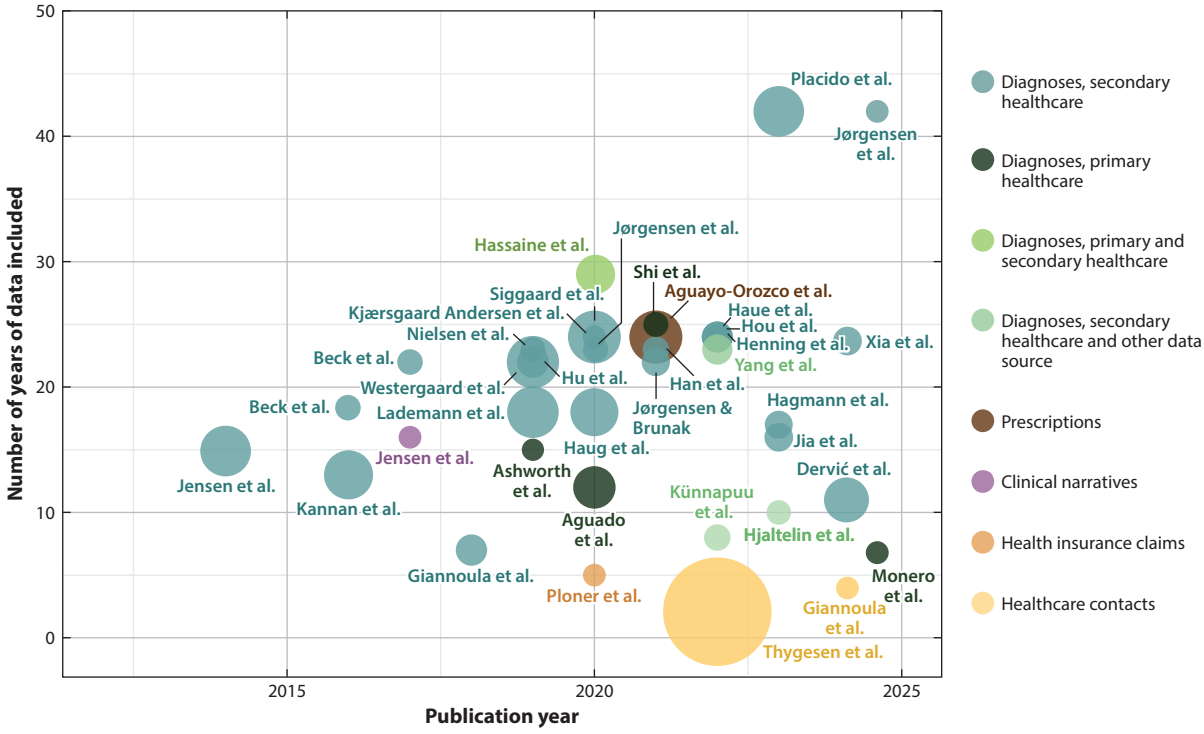


Figure 2

Overview of selected literature on disease trajectories based on data from European countries. The axes denote the number of years of data included in the trajectory analysis and the publication year. The size of each circle denotes the number of patients included in the study, and the colors correspond to the type of data used for trajectory analysis. For details of each study, see **Table 1**.

Disease trajectory studies vary greatly in the number of individuals included. The studies covered here range from fewer than 6,000 (32, 33) to almost 57 million patients (34). The follow-up periods for which disease trajectories are generated vary from almost 2 years (34) to more than four decades (20, 33). Most of the approaches create disease trajectories using diagnoses from either the primary or secondary care sector. Primary care providers are typically the first point of contact for patients, often before they are fully diagnosed, and report information on early symptoms and risk factors that could be indicative of underlying diseases later to be diagnosed more precisely. The use of primary care data for trajectories thus has the advantage of uncovering earlier symptoms compared to data from the secondary care sector. However, due to the organization of the primary care sector in many European countries, where primary care data are distributed and less centralized compared to hospital data, most of the studies in the current literature are based on diagnostic codes from the secondary care sector. Only few studies take advantage of healthcare events other than diagnoses in the creation of disease trajectories, for example, prescriptions or events extracted from clinical narratives (**Figure 2**; **Table 1**). Population-wide prescription datasets have been used to construct trajectories to describe how individuals' medication usage evolves over time and thereby reveal drug shifts, pinpointing drugs that may be ineffective or unnecessary and suggesting alternative initial treatments different from the standard first-line therapies (13). Clinical narratives offer an information-rich data source, and extracting terms from them allows the inclusion of many more variables than just what might be present in the structured parts of EHRs (35). Different text and data mining techniques can be explored to extract all the phenotypic

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information, but these methods can be time-consuming, and the end result in terms of trajectories will only be as good as the validity of the events extracted from the clinical narratives. Jensen et al. (36) used clinical narratives from EHRs and extracted diagnoses, symptoms, disease events, admission type, medication events, and surgical procedures, which they included in disease and event trajectories of cancer patients. Hjaltelin et al. (37) compared trajectories of symptoms from clinical narratives with diagnoses from structured EHRs for pancreatic cancer patients. They furthermore found that many diagnoses belonging to chapter 18 of the International Classification of Diseases, 10th Revision (ICD-10), "Symptoms, signs and abnormal clinical and laboratory findings, not elsewhere classified," consisting of general medical complaints like back pain and elevated blood glucose, are mainly discovered in the clinical narratives and not coded in the structured part of EHRs (35, 37).

Additionally, several studies have attempted to combine different, structured, health-related events into trajectories. Yang et al. (38) created biomarker disease trajectories consisting of a biomarker and disease trajectory (biomarker → disease → disease) to reveal clinically relevant subgroups among women, which may shed light on promising targets for early detection and prevention of these diseases. From claims data, Ploner et al. (39) used diagnoses, medication, and procedures to create a temporal trajectory to estimate the variables with high correlation to severe chronic obstructive pulmonary disease (COPD). Prescriptions, diagnoses, and claims have also been combined to study disease trajectories in general and specifically those of type 2 diabetes mellitus patients (40). One study used a nondisease phenotype as the starting point for a disease trajectory, namely individuals with high neuroticism, a core personality trait (41).

From Structured Healthcare Data to Disease Trajectories

For healthcare data to be summarized as disease trajectories, ontologies are needed. Medical ontologies provide a structured framework for annotating and organizing medical information, including concepts, terms, relationships, and classifications within the healthcare domain (42). These ontologies provide a common language for researchers and clinicians to precisely classify patients using computational tools. Even though disease ontologies provide a common language, they can cause challenges in creating, comparing, and validating disease trajectories over long periods of time when the ontological schemes used in routine care have been changing. Several different ontologies are used to define disease trajectories, including the ICD for diagnoses and symptoms, the Anatomical Therapeutic Chemical Classification System for medication, and the Observational Medical Outcomes Partnership for a range of observational events. Some of the most used ontologies are described below.

Ontologies Used for Secondary Care Sector Data

Most of the studies utilizing diagnoses from the secondary care sector use the ICD terminology. Originally the backbone of producing comparable mortality statistics that date back as far as 1893, the ICD system has developed into an important resource in healthcare systems that are increasingly influenced by advances in information technology (43). The very first ICD edition, the "International List of Causes of Death," included 179 causes of death that were consolidated from lists being used in individual countries (44). The ICD system is now managed by the World Health Organization and provides a standardized way of categorizing diseases and health conditions; it has undergone multiple revisions and updates over the years to adapt to changing medical knowledge and advances. It employs a hierarchical structure that categorizes diseases into a systematic framework consisting of chapters, sections, categories, subcategories, and codes. One inherent advantage of the ICD terminology is that it is generalizable across many different healthcare systems,

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as it is used in many countries (42), even if most countries have adapted and amended the international version nationally. At the chapter level there will be a high degree of information loss, causing difficulties in discovering clinically meaningful associations in many cases. In contrast, at the most detailed level, diseases are often so specific that in most cohorts there will only be a few patients following longitudinal patterns at that level, reducing the statistical power of the analysis. A comparison between disease networks utilizing the three-character disease codes and a collapsed version, a two-character coded network, revealed that the collapsed network is not recommended to directly make biological or clinical inferences (19, 27). Thus, most of the studies applying ICD codes to investigate disease trajectories use the three-character disease codes, for example, "I10: Essential Hypertension."

Ontologies of the Primary Care Sector

Several ontologies are used in the primary care sectors, complicating comparisons between studies. The International Classification of Primary Care (ICPC) is a system for classifying and coding health-related issues encountered in primary care settings that is used in the Netherlands and Spain (45, 46), among other countries. The ICPC focuses specifically on issues that are typically addressed by primary care physicians. Disease trajectories have been created using both ICD codes and ICPC-2 codes and using the ICD-10 system resulted in a multimorbidity network with twice as many diseases included and almost three times as many links between diseases as using ICPC-2 codes only (47). Mapping of disease codes is necessary when data are available from two different systems, as is the case with primary care and hospital data in the United Kingdom (48). In the United Kingdom the primary care system uses Read codes as the standardized medical coding system (49); these were mapped to ICD-10 codes as the hierarchical structure is more convenient for data mining (48). The Danish National Health Service Register (NHSR) combines health service codes from most fields within primary care including general practitioner, physiotherapy, dentist, psychology, chiropractic, etc. (50). In a recent study Bojesen et al. (51) used the NHSR and general practitioner laboratory values to predict pancreatic cancer onset in a real-time prospective machine learning (ML) setup.

METHODS FOR COMPUTING DISEASE TRAJECTORIES

Several methods have been used to compute disease trajectories, as shown in **Table 1**. Here we discuss some of the methods explicitly, starting with the simpler methods and moving to the more technical methods; we also discuss choices made by the authors in the exact computation of disease trajectories.

One of the most straightforward ways to compute disease trajectories is by simply counting the temporal order of certain diseases (32) or by establishing so-called reduced disease trajectories (20, 52). The latter include the entire disease history of a patient but with all repetitions of diseases removed. When trajectories are computed in this way, the length of the reduced disease trajectory correlates with patient age (52). Another approach went one step further, made pairwise comparisons between the patient disease histories, and only considered trajectories in which all diagnosis codes were shared in the same order by two or more patients (53).

A more common approach, which is shown in **Figure 1**, is to first identify disease pairs with a temporal association and then build longer disease trajectories from the pairs. Co-occurrences of two diagnoses (D1, D2) with a significant association are identified, meaning that for patients who have been assigned the diagnosis D1, the risk for being assigned diagnosis D2 is higher than that of a matched control group of patients (**Figure 1**). Several statistical methods have been exploited to identify such diseases that co-occur more than expected, including Poisson regression models (13),

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Cox regression models (54, 55), logistic regression models (38, 47, 56), Bayesian models (18, 57), and Markov chains (58). The strength of correlation between two diseases is often evaluated using a risk score, such as the relative risk (RR) (16, 33, 37, 59–61), the hazard ratio (54, 62), the odds ratio (38, 41, 47, 56), or association beyond chance (ABC) (57). The ABC measure is similar to the RR measure but has greater caution in assigning associations compared to RR as nearly all associations identified by ABC were also identified by RR. However, only 60–85% of the associations identified by RR were identified by ABC. Most of these discrepancies occurred for conditions with low prevalence and were some of the strongest association values found by RR (57). Several studies use a randomly selected, matched control sample based on, for example, sex, age, calendar year, or admission type. Sometimes the matched control sample can be very large; for example, Jensen et al. (16) used 10,000 control patients per case, effectively removing seasonal variation and other confounders from the pool of statistically significant, directional disease pairs identified.

The strength of the direction is then computed to indicate whether patients tend to have diagnosis D1 assigned before diagnosis D2, diagnosis D2 assigned before diagnosis D1, or there is no significant direction between diagnosis D1 and D2 (**Figure 1**). The direction has been evaluated using a binomial test (13, 16, 33, 38, 40, 54, 61–63), a Mann-Whitney test (18), between-class principal component analysis (56), and weighted association rule mining (58). Another approach to estimating directionality is using multiple linear regression models to identify statistically significant differences between age at first diagnosis (64). Yet another study tested the likelihood of developing one disease after another by calculating the causal information fraction (27). Such methods to quantify temporality can consider that some diseases might not show up in the order one expects due to disparity in prevalence. Take, for example, a specific heart disease and a symptom such as hypertension: The number of patients with hypertension greatly surpasses those with more specific heart diseases, and as a result, there will be numerous instances of hypertension occurring before the heart disease (27).

Longer disease trajectories can then be created by overlaying diagnostic associations with a direction; for example, D1 → D2 and D2 → D3 can be combined to D1 → D2 → D3 (**Figure 1**) (13, 16, 18, 40, 54, 64). Additionally, more complicated technical solutions have been exploited to characterize disease trajectories, including the gradient-boosting model (39), the general Markov framework (65), frequent item set mining (36), and nonnegative matrix factorization (48).

The majority of these studies use only the first occurrence of a disease in the disease trajectories (13, 16–18, 36, 37, 40, 48, 53, 54, 62, 63), which works well for chronic diagnoses. However, for more acute diagnoses such as pneumonia and influenza, there might be better ways of evaluating their occurrence in disease trajectories, like defining shorter time periods and thereby including a patient in more than one time period (40).

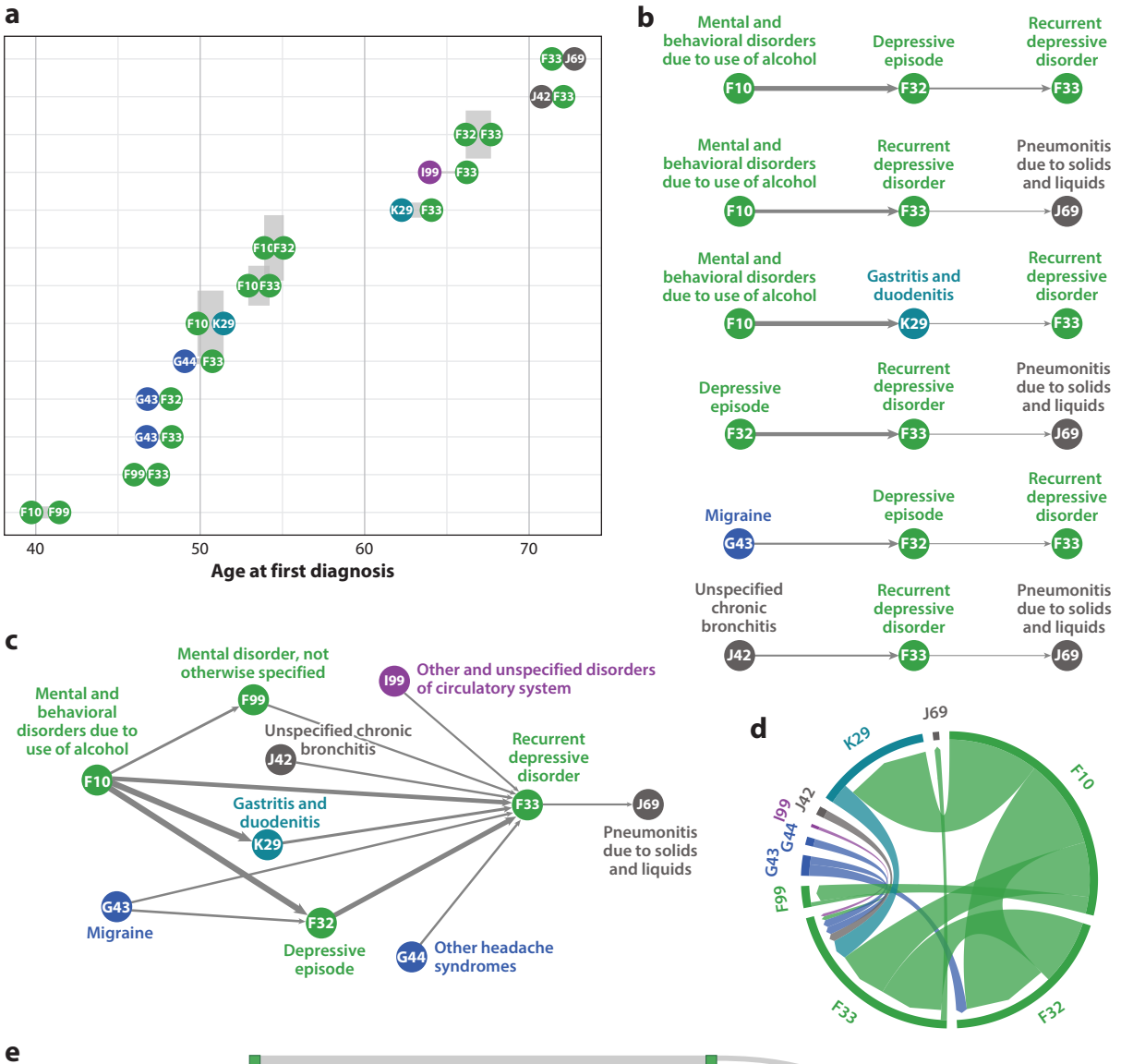
The last step for many disease trajectory methodologies is to visualize the trajectories such that they convey the results. Here, several different approaches have been used (66). Some of the more popular approaches include single, linear trajectories in either tables or smaller graphs and summaries of disease progression patterns over time in the shape of networks or diagrams (**Figure 3**).

ADDING MOLECULAR-LEVEL DATA TO TRAJECTORIES

Most diseases are at least partially heritable (67), and understanding the genetic interaction between pairs of subsequent diseases contributes to a better understanding of the shared biological pathways and the potential causality between diseases (68). Studies have also shown that disease pairs and disease trajectories can be heritable (69, 70). However, disease phenotypes not only are influenced by genetic features, but also are often a consequence of many interactions in the complex network of genes, metabolites, proteins, RNA molecules, etc. (71, 72).

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Figure 3 (*Figure appears on preceding page*)

Examples of common visualizations of disease trajectories. Disease trajectories are typically (*a*,*b*) visualized as single, linear trajectories or (*c*–*e*) summarized as disease progression paths that patients follow over time. (*a*) Visualization of temporal disease associations using the mean age at disease onset. (*b*) Simple visualization of each diagnosis step in several disease trajectories. (*c*) Network graph summarizing the disease trajectories from panels *a* and *b*. (*d*) Chord diagram summarizing disease trajectories from panels *a* and *b*. (*e*) Sankey diagram summarizing disease trajectories from panels *a* and *b*. The examples are created based on disease trajectories for F33: recurrent major depressive disorder from the Danish Disease Trajectory Browser (**http://dtb.cpr.ku.dk**). Colors correspond to chapters of the International Classification of Diseases, 10th Revision.

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It is essential to elucidate the molecular origin of diseases and comorbidities to discover novel treatments, including repurposing of drugs (73). While recent studies have generated genetic interpretations for 46% of the comorbidities identified in selected large cohorts, there is a need to improve our current understanding of the mechanisms underlying disease co-occurrence (73). For example, a disease interaction network was inferred from similarities between patients' differential gene-expression profiles (74). The authors further investigated inverse comorbidities, identified as diseases that co-occur with a lower-than-expected probability based on their individual frequencies (74). Several studies have found inverse associations among patients with central nervous system (CNS) disorders; they seem to be protected from certain cancers and particularly various types of solid tumors (75, 76). Interestingly, molecular transcriptomics evidence shows significant overlaps between genes that are upregulated in CNS disorders and genes that are downregulated in solid cancers. This opposing expression pattern could be caused by molecular germline genetic mechanisms or drug treatment for either cancers or CNS disorders (75). The knowledge about how one disease can protect against another might initiate the development of new and better treatments or drug repurposing (76, 77). Such inverse relations between diseases cannot easily or with the same confidence be extracted from medical records, since a low co-occurrence of two diseases can be a consequence of specific coding practices, misdiagnosis, differential exposures, or a limited observation window.

Hence, studies have identified comorbidities using networks of molecular information most commonly based on shared disease-related genes (78). However, only a few of these studies have evaluated temporality between diseases and created disease trajectories from molecular data based on shared disease proteins or knowledge graphs (79, 80). As these methods do not depend on prevalence of diseases, they might be useful to identify trajectories for rare diseases (80).

SEX DIFFERENCES IN DISEASE TRAJECTORIES

In many cases, biological sex has a major impact on disease development. Hormonal changes, pregnancies, lifestyle, and doctor-seeking behavior differ between men and women, leading to differences in both sex-specific and non-sex-specific diseases. Comorbidity networks change over the life course, and these changes are also affected by sex (81, 82). Some studies have investigated the difference in patient trajectories between men and women in a more limited manner, as exemplified by one study that investigated the risk factors of chronic heart disease and found more multimorbidity in women compared to men (88.47% versus 84.36%, respectively). However, females were older when diagnosed with chronic heart disease (mean age 73.0 versus 67.5) (65). Women have also been found to be at an increased risk of endocrine, skeletal, and digestive diseases (38), and the risk for cardiovascular diseases increases significantly more in females than in males when diagnosed with diabetes, hypertension, and metabolic disorders (52).

One comprehensive study looked at diseases across 18 different ICD-10 chapters and compared individual diseases, comorbidities, and trajectories between men and women (18). On average, women received their first hospital diagnosis at a later age than men, meaning that females

tended to have more disease-free years. For certain diseases it is also possible that less detailed workup or other factors delayed the diagnoses. Interestingly, when investigating temporal comorbidities, some diseases showed opposite temporal relationships in men and women, which highlights the importance of investigating sex-stratified disease trajectories (18).

APPLICATIONS OF DISEASE TRAJECTORIES

Disease trajectories have a diverse array of applications in the realm of medical research. They play a crucial role in validating and confirming known associations. In cases where smaller studies generate new hypotheses that have yet to be confirmed, trajectories based on population-level data may hold substantial statistical power for validation. Moreover, trajectory studies can extend beyond validation to enhance existing knowledge in relation to temporality. For example, certain comorbidities linked to skin diseases were already known, but the temporality of these associations remained uncertain; this could be resolved by investigating temporal disease trajectories (83, 84). By elucidating whether the skin disease preceded the comorbidity or vice versa, trajectory studies can refine existing knowledge, providing valuable insights for future research. Using large datasets, it is possible to discover clinically relevant associations that may not have been noticed by individual clinicians but might become apparent only when data are aggregated on a broader scale comprising millions of patients (27, 85). **Figure 4** showcases diverse examples of the range of ways in which disease trajectories can be used that are further exemplified below.

Risk Identification and Early Detection

A major advantage of disease trajectories is their potential to comprehensively shed light on risk factors associated with specific diseases (39, 86). As disease trajectories consider the temporal dimension, they represent a tool to identify diseases or events happening before a disease of interest and thereby drive understanding of what factors increase the likelihood of a particular disease or health condition.

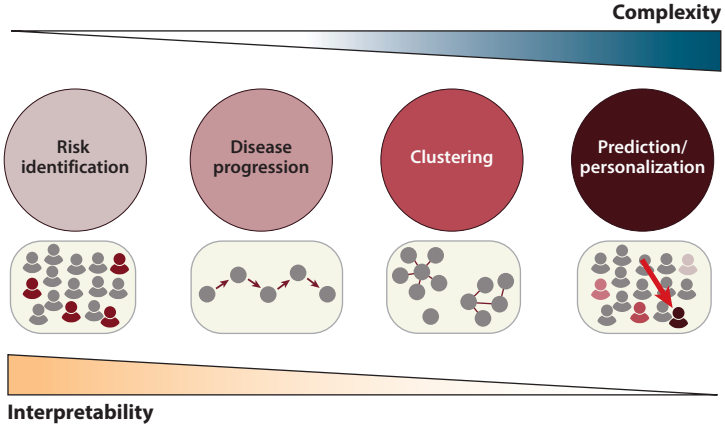


Figure 4

Examples of applications of disease trajectories. Common applications of disease trajectories include identification of risks for a specific disease to aid early detection and understand disease progression pathways of multimorbidity. Clustering can stratify patients into homogeneous subgroups, while prediction applications often use disease trajectories as input to predict future events or estimate mortality. The interpretability decreases when going from left to right, with prediction applications often being the most difficult to understand. Conversely, the complexity of the applications increases from left to right, with prediction applications often being the most complex.

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The risk factors or early signs occurring before a disease could be used for identification of high-risk patients, early detection, or proactive management of disease, making it possible to optimize treatments or even prevent certain diseases (39). Additionally, disease trajectories do not necessarily focus on one disease or one risk factor at a time. They can be used to understand the importance of the co-occurrence of multiple diseases and networks of risk factors (86). One example is a study that identified sex-specific conditions preceding COPD using disease trajectories (56). The study suggested that knowing the sex-specific conditions before COPD can enhance our comprehension of the disease's initial presentation and help uncover significant factors linked to the future development of COPD. Following a disease trajectory has often been linked to a higher disease burden and higher mortality compared to patients not following a systematic, progression disease trajectory. This was the case in a study of sepsis trajectories, in which patients approached the disease via three very different paths compromising diabetes, anemia, and alcohol abuse (59). Following the path with alcohol abuse before sepsis led to a higher sepsis mortality compared to the paths of diabetes and anemia, but following all three paths at the same time led to the highest sepsis mortality (59). Similarly, disease trajectories have found associations between osteoarthritis, heart failure, and ischemic heart disease, which have been described in detail (64). The study presents evidence that a temporal association between osteoarthritis and heart failure depends on the diagnostic order: osteoarthritis, chronic ischemic heart disease, and heart failure. The study did not identify a temporal association between osteoarthritis and heart failure only. Along the same lines, disease trajectories were used to identify populations in which osteoarthritis was most likely representing a component of metabolic syndrome and a consequence of physical activity. A recent study used 57 million patients in England to investigate COVID-19 trajectories using primary and secondary care data (34). The utilization of primary care data enabled the researchers to capture mortality outcomes for the COVID-19 trajectories based on diagnosis and vaccination information in the primary care sector.

Illuminating Disease Progression

Disease trajectories can be used to gain insights into disease progression and aid in the understanding of the evolution of multimorbidity (87). Some diseases, or combinations of diseases, might lead to faster progressions. One way of investigating disease progression using disease trajectories is to discover trajectories that diverge—that is, trajectories that start in similar ways but at some point take different directions. For example, for men diagnosed with sleep disorders at a young age (20–39 years old), trajectories typically diverge into metabolic diseases or movement disorders (60). At the time of sleep disorder diagnosis the two groups had the same burden of diagnoses, but 10 years later, patients who diverged into movement disorders had significantly more diagnoses on average than the patients who diverged into the metabolic trajectories (5.1× and 3.9×, respectively) (60).

Revealing Underdiagnosed and Misdiagnosed Diseases

An intriguing application of disease trajectories is to investigate conditions that may have been under- or misdiagnosed. By discovering patterns in the order in which diseases occur, it might be possible to identify when some diagnoses are missing or if some patients do not tend to follow typical disease patterns. One study took advantage of this and identified common disease trajectories for COPD patients that most of the patients followed (88). However, a group of COPD patients with low similarity to these common trajectories had a much higher mortality after the COPD diagnosis compared to the general COPD patients. Using deep EHR data, researchers observed that several laboratory values were more similar to the values of lung cancer patients, indicating that some of these patients might be misdiagnosed with COPD and underdiagnosed with lung cancer, which would explain their higher mortality as the cancer remained untreated (88).

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Cluster Analysis and Stratification

One popular application of disease trajectories is stratification of patients with shared trajectory patterns based on cluster analyses (31). Previous findings suggest that identifying clusters of multimorbidity can identify homogeneous groups of patients with similar prognoses, which can support clinicians and healthcare systems in tailoring care and treatments to better suit the needs of these specific groups (87).

Several studies have computed disease trajectories for a patient group and afterward clustered the trajectories to discover the main multimorbidity patterns (16, 27, 33, 52, 53, 89). From more than 5 million patients in Austria, researchers clustered disease trajectories into 132 different multimorbidity patterns (52). Each of the clusters was defined by a set of diseases that patients had been diagnosed with and a set of diseases that the patients had not been diagnosed with. The authors correlated each cluster with mortality and found that clusters with high mortality often included hypertensive diseases, heart diseases, cerebrovascular diseases, and malignant neoplasms (52). In another study, dynamic time warping was used to cluster sex-stratified disease trajectories according to temporal characteristics (53). The authors identified 199 highly populated clusters for males, with the two most populated clusters comprising trajectories with diseases of the respiratory system and diseases of the circulatory system, respectively. For females, the authors identified 164 clusters, with the largest clusters comprising complications of pregnancy, childbirth, and the puerperium and diseases of the respiratory and circulatory systems and different combinations of them (53). Similarly, Jensen et al. (16) identified 15 clusters among disease trajectories from the Danish population and identified gout and COPD as crucial diagnoses that are important for further disease progression and therefore essential to diagnose early to reduce the risk of unfavorable outcomes. Disease trajectories have previously been stratified using symptoms as an independent variable, which implies that symptoms play a crucial role in establishing a subsequent diagnosis (90). Disease trajectories for patients with depression identified three main clusters compromising cardiometabolic disease, chronic inflammatory disease, and tobacco abuse (54).

Prediction and Personalization

The sequence of diseases represented by disease trajectories can additionally be used for prediction tasks, like forecasting comorbidity, disease burden, survival, and other outcomes (52). This can shed light on risk assessment, disease prognosis, and treatment planning by guiding clinical decision support tools (87). Especially within the field of cancer research, disease trajectories have been used for prediction. For example, text-mined disease trajectories predicted 80% of cancer patient events ahead of time, enabling optimized and personalized cancer treatments (36). Similarly, disease trajectories based on diagnoses have been used to predict which patients are at higher risk of developing pancreatic cancer (20).

TRAJECTORIES AND MACHINE LEARNING

Artificial intelligence (AI) models have proved to be good solutions for learning longitudinal patterns within disease trajectories; examples include predicting adverse events in a hospital setting and readmission or mortality in the intensive care unit (19, 91). Trajectories including continuous features such as vitals sign measurements are classic examples of time series data that ML and deep learning (DL) models can learn from. While many traditional models do not take the temporal aspect into account, DL can retain previous events in memory and make predictions, putting different weights on events that occurred previously in time and are associated with the outcome. These kinds of temporal trends are not easy to detect in data unless assumptions are

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already made about such variations, like seasonality changes in laboratory test measurements (92). Although classic ML models like random forest and boosting algorithms can be used for regression tasks with time series, deep neural networks have proved to be particularly well suited for these situations. The two main types of approach that neural networks use to learn temporal patterns are recurrent and attention mechanisms. In the first, the model can remember previously presented input information because the nodes in the network generate an output that is conditioned on previous time points. The attention mechanism allows the model to learn patterns in sequences by weighting each element depending on its contribution to the outcome. Both models have proved to outperform more classic approaches when learning from disease trajectories.

Studies have used ML and DL to integrate longitudinal information from EHRs, which are not limited to disease trajectories but often include medication, events from clinical narratives, etc. (93–95). The predictive value of disease trajectories will likely be exploited even more in AI models in the near future to reveal novel patterns within patient datasets (8, 96).

CHALLENGES OF DISEASE TRAJECTORIES

There are challenges related to various aspects of disease trajectories, including the data, the methods, and the applications, that must be considered and adjusted for each specific study. Some of these challenges might be overcome by adapting the study setup, while others, such as issues related to the quality of data, are more difficult to fully solve.

Quality of the Data Foundation

The data foundation for disease trajectories is often based on elements from EHRs or specific registries. These data are collected for routine clinical care and billing purposes, and as a result there can be missing, imprecise, and erroneous data, which makes it challenging to find the ground truth (97, 98). Missing data can be due to a lack of information, where prescriptions or diagnoses from the general practitioner are not available, or due to incompleteness where data are not recorded. For example, diagnoses not leading to financial reimbursement, e.g., overweight, can be underreported (52, 99). There are techniques to impute missing data, but depending on the amount and type of data missing and the method used for imputation, these techniques can introduce bias and thereby affect the results, and so such analyses should be handled with care (100, 101). Some of the challenges with diseases not being coded can potentially be tackled using text mining techniques to extract information from the clinical narratives if these are available (36). Studies have shown that clinical notes are enriched for certain diseases compared to using diagnosis codes from registries (35, 37). However, text mining requires a lot of additional work, and while registries often contain a relatively homogeneous recording of codes across disease areas, clinical narratives vary a lot in different domains of disorders. It takes time and resources to understand the unstructured variation in clinical narratives and appreciate how this variation can be handled uniformly in the context of multimorbidity (36).

Imprecise data occur when the data are valid but not specific enough to capture the information needed to build trajectories (100). This can happen in the specification of disease subtypes; for example, many patients diagnosed with dementia do not have a further specification of the subtype of dementia (e.g., Alzheimer's or vascular dementia), which can be crucial for understanding disease pathophysiology, diagnosis, and treatment optimization longitudinally. A related problem is mixed etiologies, in which several subtypes of a disease (e.g., dementia) can coexist in the same patient (102). In some disease areas these mixed etiologies are known and emerging diagnosis codes may exist covering them (e.g., mixed dementia coding in ICD-11), while in other areas like

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type 2 diabetes they are still being characterized within research projects and thus are not generally available in historical routine care data (103–105).

Furthermore, the severity of diseases cannot always be captured well using diagnosis codes. In coding systems like ICD, some diagnoses indicate a certain degree of severity, but in UK Biobank data, for example, it is not always possible to appreciate the severity of the disease (55). Detailed diagnostic information is often not available although it can be crucial to understand the correctness of a diagnosis; for example, test results may be needed to confirm a certain diagnosis (56).

Even though the data resources used to create disease trajectories are often considerable and population-wide, there are biases to be aware of. For example, disease data from hospital records often represent a more severe disease phenotype compared to what exists in the general population (106, 107). On the other hand, data from cohort studies like the UK Biobank may include a healthier subset of the population (108). Furthermore, in the UK Biobank some of the disease information is self-reported, which can be an advantage as diseases throughout the lifetime of the individual can be reported more comprehensively. However, if participants do not remember all their diseases or the sequence in which they were diagnosed, it might influence subsequent analyses. Several large-scale studies benchmark the validity of healthcare data; in one case they concluded that the diagnoses from the Danish National Patient Registry are consistently accurate (109, 110). Similarly, the accuracy of diagnostic coding in the United Kingdom has been improved, resulting in an overall accuracy that for many studies is sufficiently robust (111).

Differences in Coding Practices

As healthcare data are also collected for financial reimbursement and statistical purposes, there might be biases in coding practice that need to be considered when using the data for research (8, 112, 113). Additionally, coding practices differ between hospitals and physicians; thus, patients with identical phenotypes might be assigned different codes (99). The same conditions might be coded differently using congruent codes (e.g., depressive disorders coded as either F32 or F33 in ICD-10), and coding rules might slightly differ between two systems (e.g., the use of codes for pneumonia versus bacterial pneumonia) (56). The correctness of diagnosis codes depends largely on the skills of the healthcare workers and the time available to review the patient record data (56).

Besides the differences in coding practice between individual healthcare workers and hospitals, the meaning of individual diagnosis codes may also change. The ICD-10 code E11 was originally defined as "noninsulin dependent diabetes mellitus" but this was changed in 2014 to "type 2 diabetes mellitus," which could potentially affect how healthcare workers use the diagnosis code (47). As medicine evolves continuously, diagnostic criteria and methods change over time, contributing to differences in coding practices (65, 99, 113). For example, the lowering of the threshold for diagnosing diabetes from a fasting glucose level of $\geq$140 mg/dL to $\geq$126 mg/dL (114) and the inclusion of troponin as a biomarker for diagnosis of myocardial infarction caused an increase in the number of individuals diagnosed with the two diseases (115). Additionally, better high-resolution diagnostic technologies, advanced imaging, and greater access to tests through self-testing technologies contribute to changing coding practices (116).

Perspectives on the International Classification of Diseases, 11th Revision

In 2018 ICD-11 was released; it includes over 100,000 medical diagnostic index terms encompassing all possible injuries, diseases, and mortality causes (**https://icd.who.int**). The progress in understanding underlying molecular mechanisms and clinical phenotypes of diseases has contributed to the distinction between disease and subtypes of disease, and this is reflected in the

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substantial increase in the number of codes available in each revision of the ICD. The overwhelming amount of more specific phenotypes, including combinations of phenotypes, can be good for stratifying patients correctly and understanding additional levels of disease severity, but it may also be important to consolidate codes at a lower level to ensure the codes populated with enough patients so statistically significant results can be obtained.

Heterogeneous Temporal Data and Challenges with Cross-Country Validation

Often disease trajectories are created based on a data foundation from a single area or country. Very few studies have validated disease trajectories with populations from other countries, and in these few cases, they often replicate their own disease trajectories based on public summary data (16, 40, 56, 62). Many disease trajectories could be replicated within the public summary data, but some discrepancies were found, which likely could be explained by differences in disease frequencies, specificities in national modifications of disease ontologies, and differences in coding practices (40, 56, 62). Cross-country validation can be challenging as data used for calculating disease trajectories are personally identifiable and therefore cannot be shared between research groups in different countries (86, 96, 117). Hence, collaborations and the use of new methods and tools making it possible to share codes and data without releasing sensitive information are needed (8). Furthermore, new initiatives like the European Health Data Space, which aims to harness the potential of health data to improve healthcare outcomes, will eventually make it possible to analyze health data from hundreds of millions of patients. One example of cross-country validation is the discovery of early risk factors for Alzheimer's disease (118). The authors used diagnoses from both the United Kingdom and France and observed a good generalizability across the two countries, likely due to the healthcare systems being similar. Another example is the use of disease trajectories in a DL model to predict pancreatic cancer (20). The authors trained the algorithm in a Danish population and validated it in a US population from the Veterans Health Administration (VHA). The performance was not as good in the US population as in the Danish validation data, which may be partly due to stark differences in multimorbidity levels across the populations (more severe in the VHA data), but also to well-known differences in coding practice, with more frequent coding in the United States. Ultimately, generalizability of disease trajectories is strictly dependent on the data foundation, for example, public versus private healthcare sector, population-wide versus subset of a population, and different purposes of data collection. However, cross-country validation of disease trajectories is essential to understand which multimorbidity progression patterns are unique to local or national healthcare systems and which are due to biological mechanisms or differences in exposure and are generalizable globally (31, 40).

Most disease trajectories studies are based on data from high-income countries (31). The similarities of disease trajectories based on data from low- and middle-income countries might be even smaller, but this has not been fully elucidated due to a lack of data availability. Many low- and middle-income countries might not have the possibility to gather structured hospital data necessary for large-scale temporal studies or replication of previously developed methods (31). New AI tools together with the lack of a relevant data foundation in low- and middle-income countries will likely lead to new and larger health disparities, and there is an urgent need to focus on the possibilities to investigate disease trajectories in low- and middle-income countries and compare them more systematically (119).

Association Versus Causality in Disease Trajectories

Depending on the details of the method of creating disease trajectories, a large number of diagnosis codes often need to be investigated for possible correlation and temporal direction, leading to

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a combinatorial explosion of the number of possible disease combinations (52). A correlation does not necessarily imply causality; thus, there can be several confounding factors causing a correlation such as sex, socioeconomic status, or genetics, and hypothesis-driven experiments are needed to clarify causality (120, 121). Most disease trajectory studies are descriptive. Although definite conclusions about causal relationships therefore cannot be drawn from trajectory networks, the trajectories can identify relationships, which is a prerequisite for causality that may be investigated in future studies. This could prove particularly valuable in less-studied contexts, where disease trajectories have the potential to uncover potential risk factors and distinctive patterns, thereby generating new hypotheses for future research (39).

Clinical Implementation

One major challenge, not only for developing patient trajectories but also for many precision medicine initiatives, is moving the results generated by research into the clinic for patients to benefit. Clinical decision-making is challenging, and many of the tools created based on patient trajectories will be very relevant to inform about potential interventions (36). Some studies identify risk factors for clinicians to be aware of when handling a certain patient group. For example, one study found type 1 diabetes mellitus, acute myocardial infarction, and pneumonia often occur in patients with newly diagnosed hidradenitis suppurativa and recommends that clinicians be aware of these risks (84). In other cases, disease trajectories are used in more complex prediction tasks that cannot be immediately implemented in the clinic. For example, prediction of mortality in the intensive care unit based on disease trajectories was shown to outcompete other current acute data-based methods for estimating mortality risk (19). However, before these types of applications of disease trajectories can be used in the clinic, results may need to be confirmed in randomized clinical trials.

FUTURE DIRECTIONS AND CONCLUDING REMARKS

The concept of disease trajectories is essential as more patients live with multiple morbidities and receive multiple medications. Disease trajectories bring together aspects of temporal information and have proved to be predictive in several studies. Additionally, the temporality has proved to be useful for prediction of both comorbidity burden and mortality. Disease trajectories have several applications, including risk identification, patient stratification, and outcome prediction. The increase in EHR datasets and available patient information together with integration of AI techniques will likely increase the number of disease trajectory studies in the future. One major gap in disease trajectory studies is cross-country validation, which future studies should focus on. Knowledge about whether the discovered trajectories are an artifact of the specific healthcare system or whether they reflect an actual biologic signal is essential for future studies.

DISCLOSURE STATEMENT

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